DAA questions (2m)

1.what is an alg design technique.give an ex.

An **algorithm design technique** is a general method or approach used to create efficient algorithms to solve a problem.

Example:
Divide and Conquer – This technique divides a problem into smaller subproblems, solves them independently, and then combines their results.

👉 Example: **Merge Sort** (splits the array, sorts each half, then merges them)

2.define alg analysis.

Algorithm analysis is the process of evaluating an algorithm's efficiency in terms of **time** and **space (memory)** required as the input size increases.

3.list and explain characteristics of an alg

Characteristics of an Algorithm:

1. **Input** – It should take zero or more inputs.
2. **Output** – It should produce at least one output.
3. **Definiteness** – Each step must be clear and unambiguous.
4. **Finiteness** – It must terminate after a finite number of steps.
5. **Effectiveness** – Each step should be simple and executable in a finite time.

4.what is emperical analysis of alg

Empirical analysis of an algorithm is the method of evaluating an algorithm by **implementing it and running it on a computer**, then measuring its **actual execution time and memory usage** for different inputs.

5.list the methods of computing the time efficiency of an alg

Methods of computing the time efficiency of an algorithm:

1. **Analytical (Theoretical) Method** – Calculates time using mathematical analysis (e.g., counting operations, Big-O notation) without executing the program.
2. **Empirical (Experimental) Method** – Measures actual running time by implementing and executing the algorithm on a computer.

6.what is time complexity of an alg.

Time complexity of an algorithm refers to the amount of time taken by an algorithm to run/complete a program/task as the input size grows.
It is usually expressed using notations like **Big-O**.

7.what is space complexity of an alg.

space complexity of an algorithm refers to the amount of space taken by an algorithm to run/complete a program/task as the input size grows.

8.write the order of growth of functions

Order of growth of functions (in increasing order):

[O(1) < O(\log n) < O(n) < O(n \log n) < O(n^2) < O(n^3) < O(2^n) < O(n!)]

It shows how the running time increases as the input size grows.

9.define brute force approach

Brute force approach is a simple algorithm design technique that **tries all possible solutions** to a problem and selects the correct one.

It is easy to implement but usually **inefficient** for large inputs.

9.what is exaustive search.

Exhaustive search is an algorithmic technique that **examines all possible candidates (solutions)** to find the correct or optimal one.

It guarantees a correct solution but is usually **time-consuming and inefficient** for large problems.

11.what is the worst case analysis of linear search.

Worst-case analysis of Linear Search:

In linear search, we **check each element one by one** until we find the target or reach the end.

- **Worst case:** The element is **not present** or is at the **last position**.
- **Number of comparisons:** (n) (for an array of size (n))
- **Time complexity:** (O(n))

This means the search may need to examine **all elements** in the worst case.